PAEC - Philippine Aeronautical English Corpus

Aviation Training Platform

PROJECT TYPE Aviation Training Platform	**PLATFORM** Responsive web training platform
AUDIENCE Pilots, air traffic controllers, and aviation trainees	**PRIMARY FOCUS** ICAO-aligned aviation English practice

Overview

A corpus-based training and analysis platform that helps pilots and air traffic controllers improve their aviation English under ICAO standards. Built around authentic ATC communications from RPLL (NAIA), the platform delivers interactive training modules, automated dialogue analysis, and progress tracking.

Core Features

- 4 Training Modules: Scenario-Based Simulation, Readback/Hearback Correction, Jumbled Clearance, Pronunciation Drill
- Automated ICAO compliance analysis of PDF, DOCX, and text transcripts
- Phase-aware engine for APP/DEP, GND, and RAMP communications
- Web Speech API pronunciation feedback against ICAO standards
- Session history, performance trends, and per-category progress charts
- Email + Google OAuth auth with NextAuth.js 5 (JWT, bcrypt)
- PDF and CSV export of analysis results
- Backed by PAEC Corpus v3.4 - RPLL ATC recordings (Feb-Mar 2025)

Experience and Design

- Four training modes cover simulation, readback correction, clearance ordering, and pronunciation.
- Transcript analysis identifies non-standard phraseology, number errors, and safety-critical patterns.
- Phase-aware feedback distinguishes approach, departure, ground, and ramp communications.
- Session history, category breakdowns, charts, and exports make improvement measurable.

Architecture and Data

- Next.js App Router and React provide server-rendered pages and interactive training modules.
- PostgreSQL on Neon stores accounts, training sessions, attempts, and analysis history.
- Document parsers normalize PDF, DOCX, and plain-text transcripts for the analysis engine.
- Recharts presents progress trends and category-level performance in the learner dashboard.

Security and Privacy

- Bcrypt password hashing and NextAuth sessions protect account access.
- Google OAuth and verified email flows provide controlled identity options.
- Training, analysis, history, and export endpoints require authenticated sessions.
- Per-user data isolation keeps transcript analysis and progress records account-scoped.

Technology Stack

Next.js 16	Application framework and UI
React 19	Application framework and UI
TypeScript	Application language
Tailwind CSS	Interface styling and design system
PostgreSQL (Neon)	Authentication, database, or cloud data
NextAuth.js 5	Authentication and session security
Framer Motion	Animation and interaction
Recharts	Data visualization

Performance and Reliability

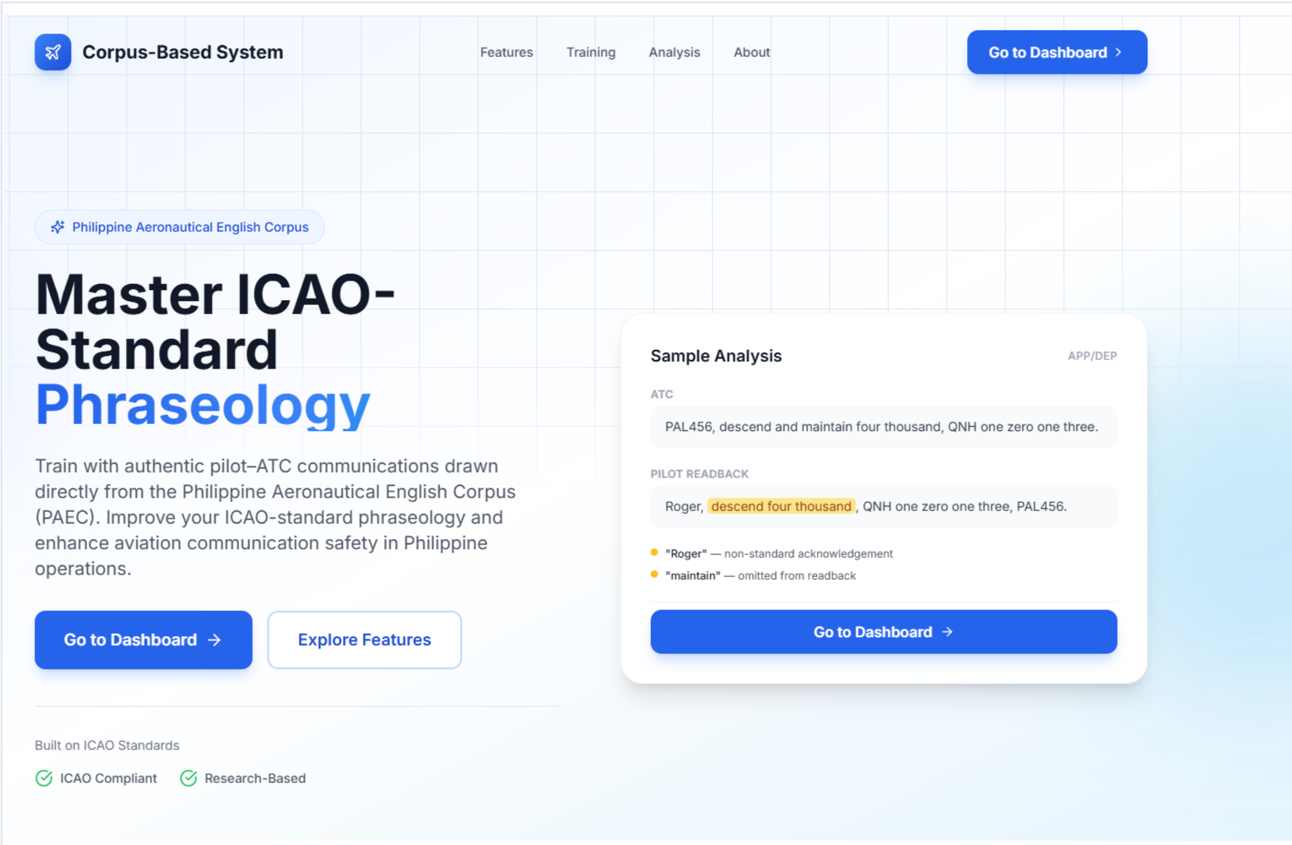
- Server components reduce unnecessary browser-side JavaScript.
- Streaming and route-level code splitting improve perceived page speed.
- Connection pooling supports efficient access to serverless PostgreSQL.
- Document parsing and chart components load only when their workflows are opened.

Platform Requirements

- Current Chrome, Edge, Firefox, or Safari browser.
- Stable internet connection for training, analysis, and progress synchronization.
- Microphone access for pronunciation practice and speech feedback.
- Chrome or Edge is recommended for the strongest Web Speech API support.

PAGE OVERVIEW

Landing Page

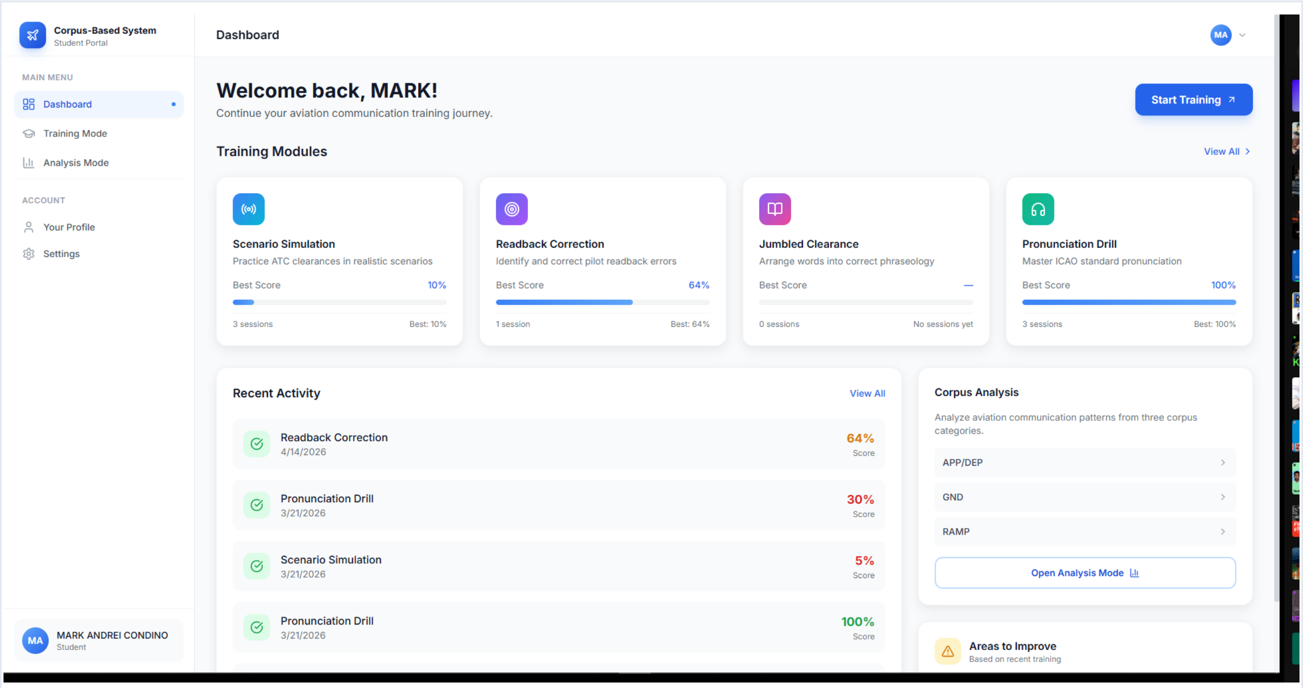


Landing Page

Interface view from PAEC - Philippine Aeronautical English Corpus.

PAGE OVERVIEW

Student Dashboard

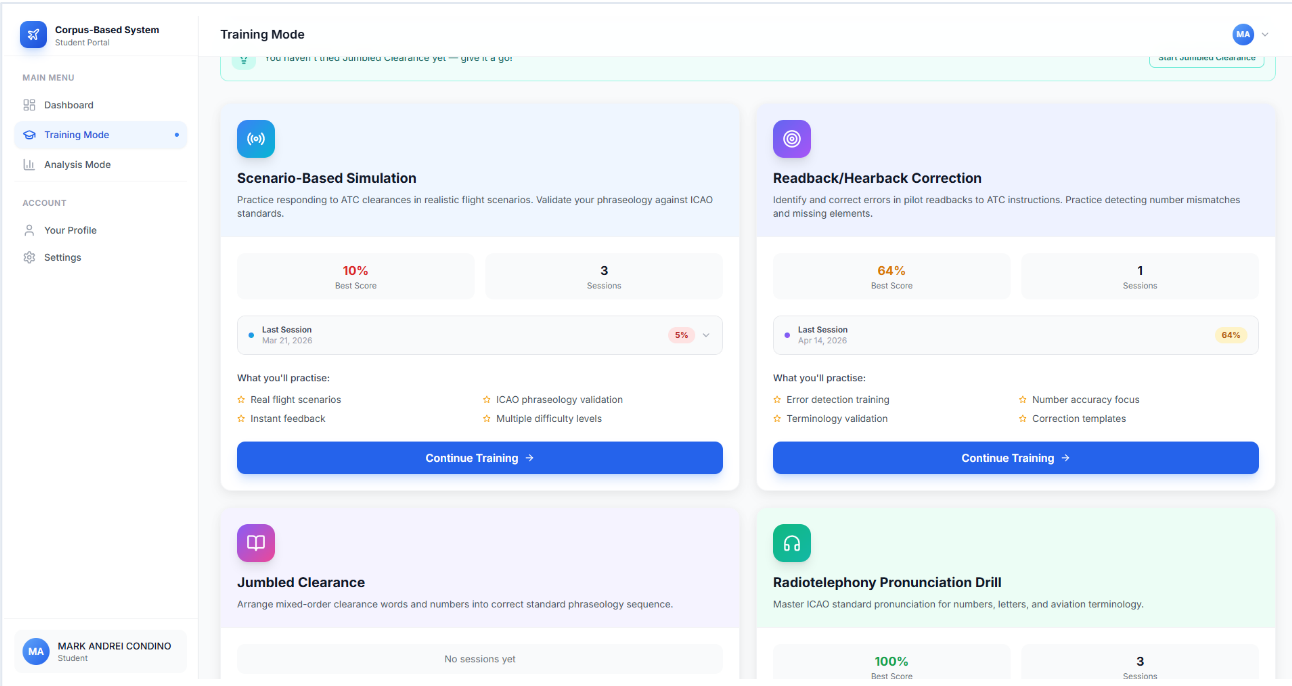


Student Dashboard

Interface view from PAEC - Philippine Aeronautical English Corpus.

PAGE OVERVIEW

Training Modules

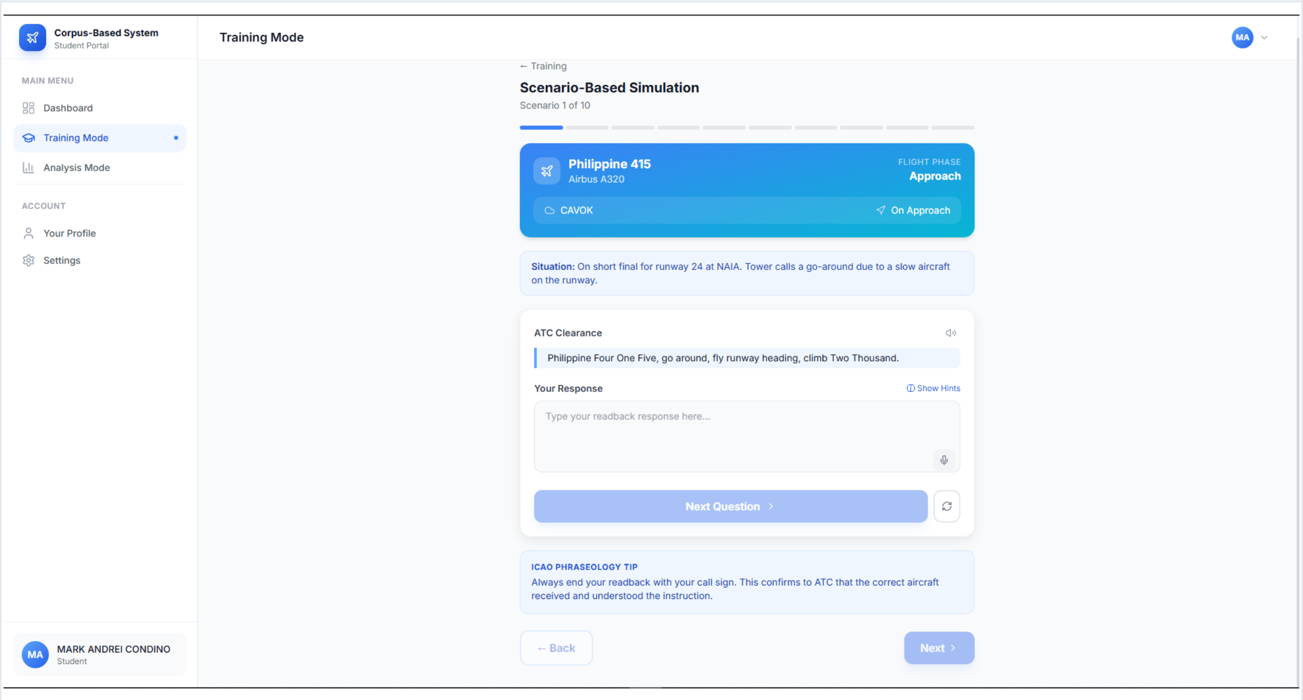


Training Modules

Interface view from PAEC - Philippine Aeronautical English Corpus.

PAGE OVERVIEW

Scenario-Based Simulation

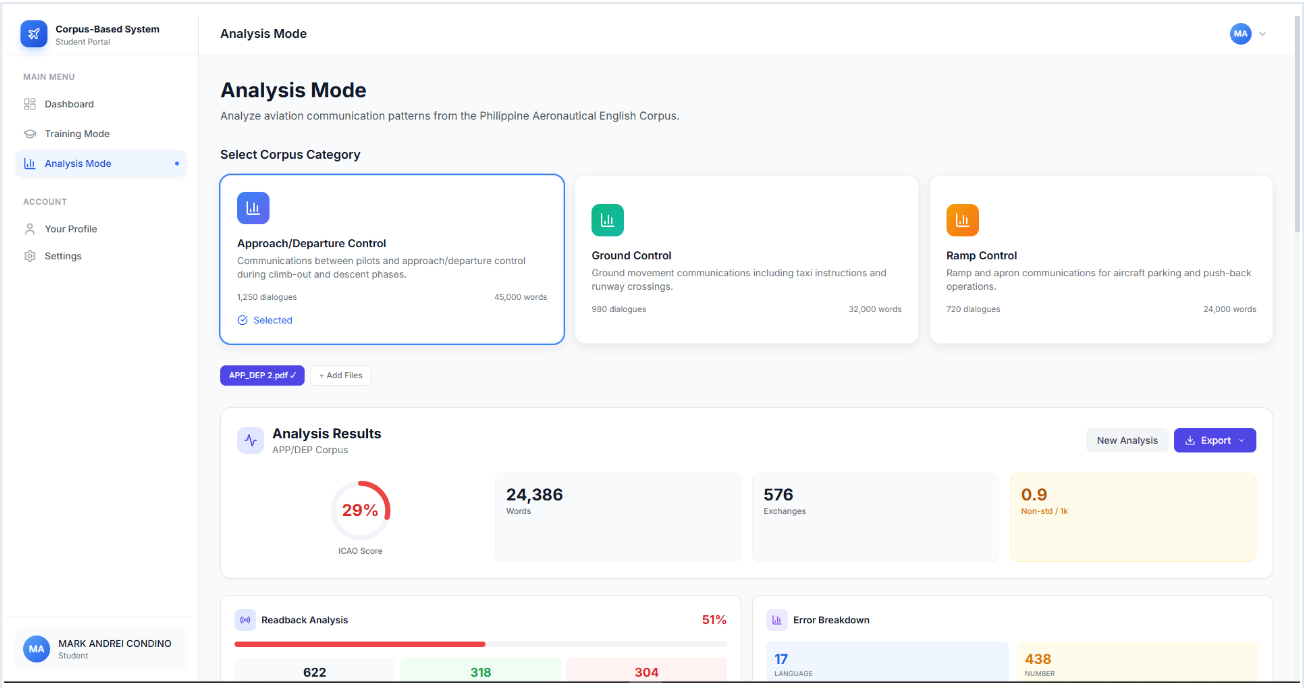


Scenario-Based Simulation

Interface view from PAEC - Philippine Aeronautical English Corpus.

PAGE OVERVIEW

Analysis Mode

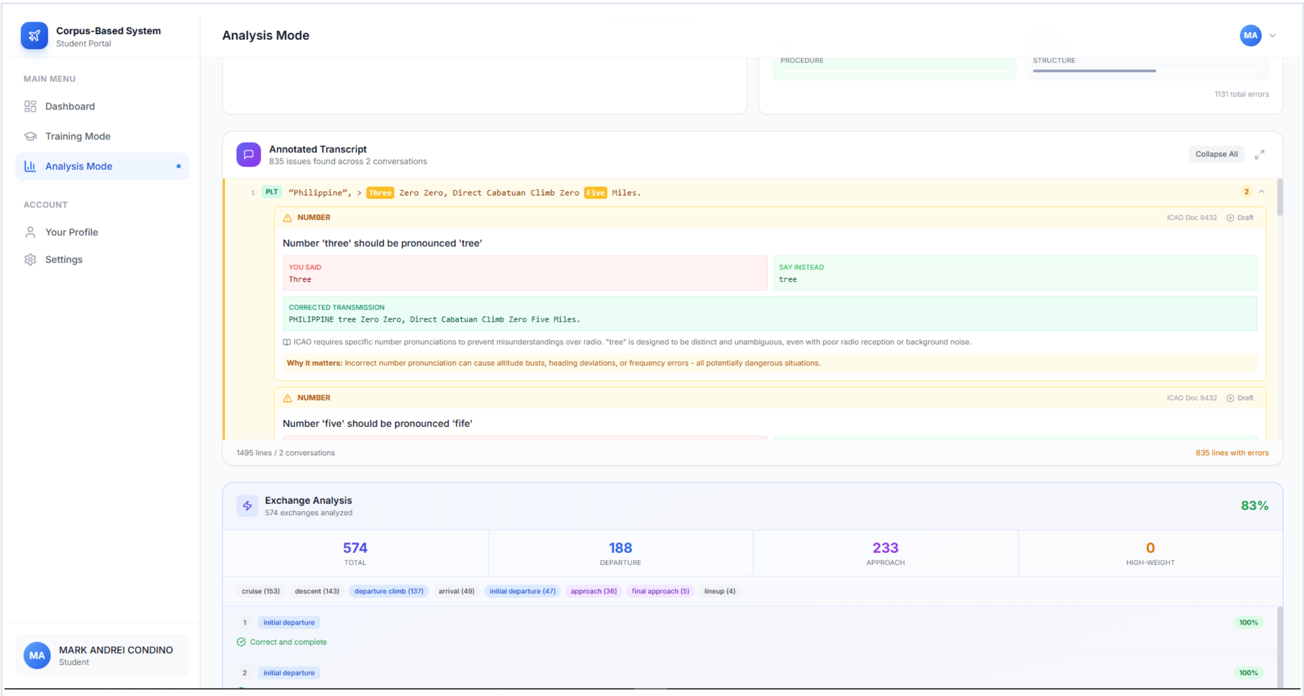


Analysis Mode

Interface view from PAEC - Philippine Aeronautical English Corpus.

PAGE OVERVIEW

Annotated Transcript

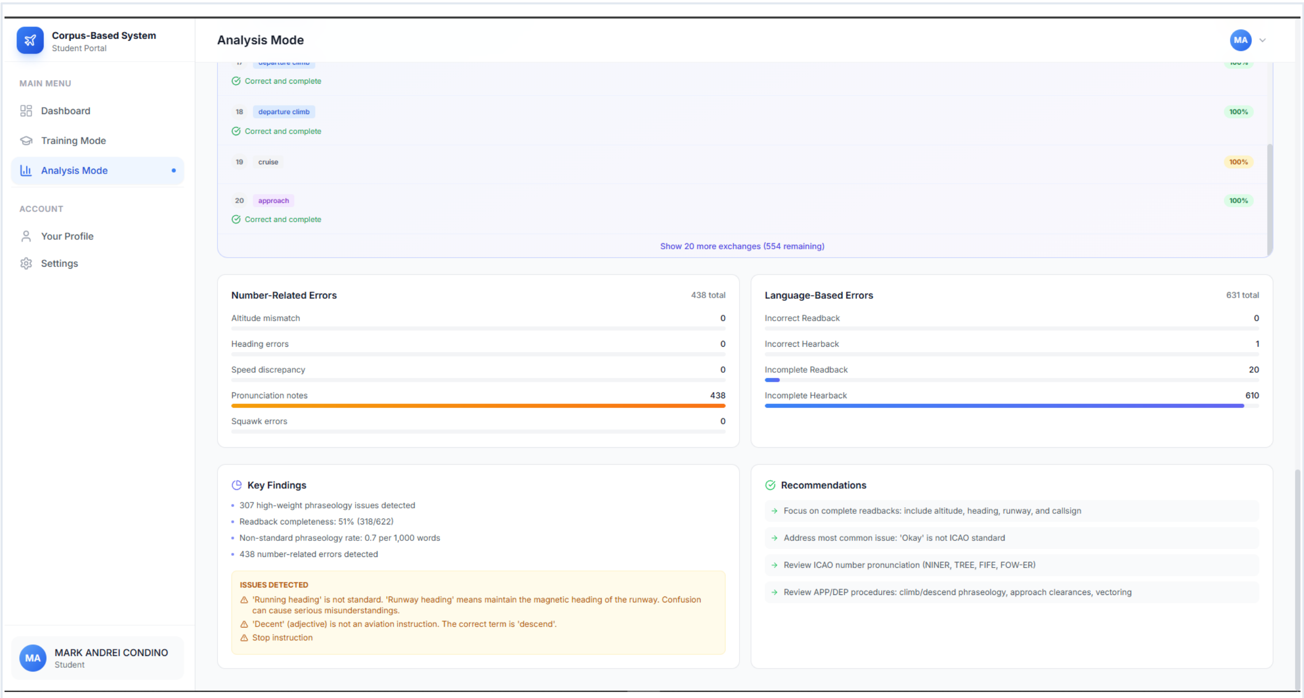


Annotated Transcript

Interface view from PAEC - Philippine Aeronautical English Corpus.

PAGE OVERVIEW

Error Breakdown & Recommendations



Error Breakdown & Recommendations

Interface view from PAEC - Philippine Aeronautical English Corpus.